ModelIt!

Student Activity Pack

The Carbon Bomb: Why Thawing Permafrost Terrifies Scientists

Modeling Arctic Feedback Loops and Climate Tipping Points

Grade 10L2 | NGSS HS-ESS2-6, HS-LS2-3 | Stability and Change

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think permafrost is and why might scientists be worried about it thawing?

2 What do you know about feedback loops — can you think of an example where a process speeds itself up?

3 Draw a diagram showing how you think carbon moves between the ground, the atmosphere, and the ocean.

4 What is a tipping point and why might one exist in the climate system?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Permafrost Temperature
- Methane Release Rate
- Microbial Decomposition
- Atmospheric CO2
- Ocean Absorption
- Vegetation Growth
- Feedback Amplification

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trajectory

Rising atmospheric CO2, continued warming

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Aggressive Cuts

Near-zero emissions growth

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Methane Burst

Sudden large-scale methane release

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Thermokarst Formation — The collapse of ground surface as ice-rich permafrost thaws, creating lakes and wetlands that accelerate further thawing and become concentrated methane emission sources

Subsea Permafrost — Frozen ground beneath the Arctic Ocean floor that formed during ice ages when sea levels were lower — now warming from both above (warmer ocean) and below (geothermal heat), potentially releasing enormous methane stores

Albedo Change — The shift in surface reflectivity as white snow and ice are replaced by dark water and vegetation in thawing regions, causing the surface to absorb more solar energy and accelerating local warming

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

There is twice as much carbon frozen in permafrost as exists in the entire atmosphere — what happens when it thaws? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the difference between a positive feedback loop and a negative feedback loop in the climate system?**

2. **Why can't Ocean Absorption and Vegetation Growth fully compensate for the carbon released from thawing permafrost?**

3. **What does your model suggest about whether the permafrost tipping point has already been passed or can still be avoided?**

4. **How does the timescale mismatch — carbon accumulated over tens of thousands of years being released over decades — create the crisis?**

5. **What are the limitations of modeling a global climate feedback with only seven components when the real system involves ocean circulation, cloud dynamics, and ice sheet behavior?**

8 STEM Challenge: Design an Arctic Permafrost Monitoring Network

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YOUR ENGINEERING MISSION

Design a monitoring system that tracks permafrost thaw rates across the Arctic and provides early warning when the carbon feedback approaches tipping point conditions.

THE SCENARIO

The International Arctic Research Council needs a comprehensive monitoring network to detect whether permafrost carbon release is approaching the self-sustaining tipping point. Your team has been hired to design the sensor network, data analysis system, and warning threshold criteria.

Engineering Design Thinking

1. WHAT TYPES OF SENSORS WOULD YOU DEPLOY — GROUND TEMPERATURE, METHANE FLUX, SOIL MOISTURE, SATELLITE THERMAL IMAGING?

2. HOW WOULD YOU POWER AND MAINTAIN MONITORING STATIONS IN EXTREMELY REMOTE ARCTIC LOCATIONS?

3. WHAT DATA ANALYSIS ALGORITHMS WOULD DETECT THE TRANSITION FROM LINEAR THAW TO EXPONENTIAL FEEDBACK?

4. HOW WOULD YOU COMMUNICATE TIPPING POINT WARNINGS TO POLICYMAKERS IN TIME FOR ACTION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists study Earth's climate systems including feedback loops, tipping points, and carbon cycles. They work at NOAA, NASA, universities, and international research organizations, earning $70,000–$140,000/year. Permafrost Researchers who specialize in Arctic carbon dynamics earn $65,000–$130,000/year.